Performance of a CDMA-based Satellite Communication System with Phase Noise

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Abstract— **In this paper, impact of phase noise on CDMA-based satellite communication system is analyzed and simulated. Satellite channel is modeled as combination of Rician fading and shadowed Rayleigh fading channels. The nonlinear Fokker-Planck method is employed to analyze the first-order PLL (phase locked loop) performance. Phase noise is modeled as Tikhonov distribution. The performance is evaluated in terms of bit error probability. The simulation results for BER performance with phase noise are shown in terms of many system parametes such as SNR, the number of users in a beam, and standard deviation of shadowing. From simulation results, it is concluded that the phase noise substantially degrades system performance of CDMA-based satellite system. This can be interpreted that detection loss (decrease of SNR) due to phase noise leads to degradation of system performance. The considerations in this paper can be applied to a satellite link design for a CDMA-based satellite system.**

I. INTRODUCTION

For universal and global communication networks, satellite system is expected to play an important role for a wide range of services. The satellite system can support numerous types of services (position finding, radio paging, data and voice communications, etc.) and various kinds of terminals (land vehicle, aircraft, marine vessel, portable terminal, etc.) [1-3].

Recently, DS/CDMA (direct sequence/code division multiple access) has attracted much attention as a promising multiple access technique for a land-mobile satellite systems and networks [4,5]. For effective and reliable communications in a DS/CDMA system, a receiver must synchronize a locally generated pseudonoise (PN) code with the incoming one. The chip synchronization of a DS/CDMA system typically consists of two steps: acquisition (coarse alignment) and tracking (fine alignment) [6]. During acquisition process, PN code uncertainty region is searched until the incoming and locally generated PN codes are sufficiently closely (typically, within a chip) aligned. PN tracking process serves to reduce the alignment error which remains after acquisition and to maintain the alignment error using feedback loops such as DLL (delay-locked loop) or TDL (tau-dither loop). For coherent detection in a DS/CDMA system, carrier synchronization is required. In this paper, under an assumption of perfect chip synchronization, we focus our attention to carrier phase synchronization problem.

With increasing interests in a wideband communication systems, the effect of carrier phase noise are becoming more important because the wideband system employs higher carrier frequencies. The phase noise can be found in many system components such as frequency multiplier and frequency synthesizer [7]. This phase noise often limits the system performance by deriving detection loss through decrease of effective SNR (signal to noise ratio). The phase noise has been known to be one of the impairment factors for digital radio communications, in the worst case, resulting in irreducible error rate performance. In a satellite communication environment, phase noise has an great impact on system performance in that a small satellite terminal (such as VSAT (very small aperture terminal)) typically uses low-cost oscillator, and it operates under low SNR (signal-to-noise-ratio) and and low data rate.

It is well known that the effect of phase noise becomes prominent in the following cases [8]: 1) low symbol SNR, 2) low data rate transmission, and 3) high carrier frequencies. In these cases, the contribution due to phase noise cannot be negligible in the analysis of system performance. For example, a satellite system typically employs high frequencies of GHz, thus it is expected that the phase noise has a significant influence on performance of the satellite system.

PSK (phase-shift-keying) is known to be an efficient modulation technique which has extensively been used in digital communications. Coherent demodulation of a PSK signal requires generation or extraction of a local carrier phase reference. So, any kind of noise associated with carrier leads to degradation of detection performance. That is, the phase noise induces detection loss resulting in degration of bit-error-rate (BER) performance. The detection loss due to phase noise at a specified BER is defined as the increase of required SNR to maintain the same BER without phase noise.

In the previous research, the effect of phase noise has been focused on coherent fiber-optic communications and narrowband single-user communication systems [9,10]. However, the analysis and methodology in an optical system can not be applicable to the radio system due to the different phase noise characteristics and the different system implementation between them [11]. So far, there has not been any approach to evaluate the effect of phase noise on a DS/CDMA system in a satellite channel. To assess the effect of phase noise in multi-access environments, multiple access interference (MAI) component should be considered in the performance analysis.

In this paper, the impact of phase noise on a DS/CDMA system is analyzed and simulated in a satellite channel. The satellite channel is modeled as a combination of Rician fading and shsdowed Rayleigh fading channels. The phase noise is modeled as Tikhonov distribution. The nonlinear Fokker-Planck method is employed to analyze the first-order PLL performance. The BER performance with phase noise is evaluated in terms of the number of users, standard deviation of shadowing, and SNR.

In Section II, models for a DS/CDMA synchronization block, satellite system, satellite channel, and phase noise are described. In Section III, interference of DS/CDMA system is modeled, and bit error probability is derived in a satellite channel. Simulation results are presented in Section IV, and conclusions are drawn in Section V.

II. SYSTEM MODEL

A. System Description

The received signal is mixed with locally generated carrier $cos(\omega_c t+\theta(t))$ where $\omega_c = 2\pi f_c$ (f_c: carrier frequency) is carrier angular frequency and $\theta(t)$ is carrier phase shift. Then, the mixed signal is integrated for a data bit duration, and it is fed into loop filter, followed by VCC (voltage controlled clock) which is used to adjust carrier phase. The feedback operation of carrier synchronization block is repeated until the perfect carrier phase is estimated, that is, $\theta(t) \approx 0$. After that, demodulator recovers transmitted data bits via integrator output in the phase dimension.

Typically, the mobile satellites use multiple spot beams, sharing the same frequency band, in each footprint to increase the system capacity. The multiple spot beams illuminate the earth to form a cellular-like layout, but with somewhat different characteristics relative to terrestrial cellular systems [2,3]. The mobile satellite multi-spot beam systems are intended to provide either a worldwide coverage for personal communication networks, or to cover large geographical areas with a high concentration of potential mobile users.

B. Satellie Channel Model

In a satellite communication system, satellites in orbit provide links between terrestrial stations sending and receiving radio signals which suffer from fading and shadowing. The envelope of received signal is given by

$$e_r = R \cdot S, \tag{1}$$

where R is fading envelope and S is shadowing envelope. In (1), R typically follows Rayleigh or Rician distributions according to absence or presence of direct component. The probability density function ($p.d.f.$) of an instantaneous power, $p = R^2S^2$, is given by [12]

$$f(p) = \int_0^{\infty} f(p|s^2\bar{p}) f_s(s) ds, \tag{2}$$

where $\bar{p} = E[P^2]$ is local mean power, $f(p|s^2\bar{p}) = \frac{R_f+1}{s^2 p} \exp\left(-\frac{(R_f+1)p}{s^2\bar{p}} - R_f\right) I_0(2\sqrt{\frac{R_f(R_f+1)p}{s^2\bar{p}}})$, R_f is a Rician factor which is power ratio of direct and diffused components, $I_0(\cdot)$ is zeroth-order modified Bessel function, and $f_s(s)$ is $p.d.f.$ of shadowing. It has been known from extensive measurements that $f_s(s)$ has a lognormal distribution and is given by

$$f_s(s) = \delta(s-1), \quad \text{unshadowing}, \tag{3}$$

$$f_s(s) = \frac{1}{\sqrt{2\pi}\rho s \sigma_s} \exp\left(-(\frac{\ln s - \rho m_s}{\sqrt{2}h\sigma_s})^2\right), \quad \text{shadowing}, \tag{4}$$

where $\rho = (\ln 10)/20$ is unit conversion constant, m_s is average of shadowing, and σ_s is standard deviation of shadowing.

In a typical satellite channel, m_s and σ_s are in the range of -7dB to -16dB and 2dB to 6dB, respectively If the direct component does not exist, $R_f = 0$ and $f(p)$ follows Suzuki-model with Rayleigh-lognormal distribution while if the direct component does exist, $R_f \neq 0$ and $f(p)$ follows Loo-model with Rician distribution [13].

III. PERFORMANCE ANALYSIS

In the performance analysis, the followings are assumed: 1) perfect power control is achieved for each satellite link by adaptive power control mechanism ($P_k = P$ for all k), 2) each satellite employs multiple beam of $B+1$, 3) gateway uses directional antenna for each satellite, 4) there are K users in each beam, 5) mobile user uses omnidirectional antenna, 6) fading is slow ensuring constant during bit duration, and 7) chip synchronizations is perfect.

A. Interference Modeling

The transmitted signal for the kth user is given by

$$s_k(t) = \sqrt{2P_k} d_k(t) c_k(t) \cos(\omega_c t + \psi_k'), \tag{5}$$

where P_k is signal power, $d_k(t)$ is data sequence with bit duration T, $c_k(t)$ is spreading sequence with chip duration T_c and processing gain $L = T/T_c$, ω_c is carrier frequency, ψ_k' is random carrier phase. It is assumed that P_k is independent of propagation delay and carrier phase, and $d_k(t)$ and $c_k(t)$ have a sequence of +1 and -1 with rectangular pulse, respectively.

Without loss of generality, let us call the first user of beam 1 'reference user'. Then, the received signal of the reference user is given by

$$\begin{aligned} r(t) &= \sum_{k=1}^{K} \sqrt{2P} R_k S_k d_k(t-\tau_k) c_k(t-\tau_k) \cos(\omega_c t + \psi_k) \\ &\quad + \sum_{k=K+1}^{B(K+1)} \sqrt{2P} R_k S_k \beta_k d_k(t-\tau_k) c_k(t-\tau_k) \\ &\quad \cdot cos(\omega_c t + \psi_k) + n(t), \end{aligned} \tag{6}$$

where τ_k is propagation delay of the kth user, ψ_k is carrier phase, R_k is amplitude of independent flat fading on each user, β_k is discrimination due to spot beam antenna patterns, $n(t)$ is AWGN with two-sided power spectral density

$N_0/2$. For simplicity, it is assumed that all users in spot beam of interest are illuminated with unity gain of isoflux antennas.

The locally generated code and carrier signal for the reference user is given by

$$l(t) = c_1(t - \tau_1)\cos(\omega_c t + \theta(t)), \tag{7}$$

where τ_1 is code phase of reference user and $\theta(t)$ is carrier phase from local oscillator. In this paper, the code synchronization is assumed to be perfectly achieved, so $\tau_1 = 0$.

The test statistic for the reference user is given by

$$Z_1 = D + I_1 + I_2 + N, \tag{8}$$

where D is desired signal component, and given by

$$D = \sqrt{2P}R_1 S_1 T d_1 \cos\phi_1, \tag{9}$$

$\phi_k = \psi_k - \theta_k$ is phase error (or phase noise) between received and locally generated signals, I_1 is multiple access interference component from the users of the same cell with reference user, and given by

$$I_1 = \sum_{k=2}^{K} \sqrt{2P}R_k S_k Y_k \cos\phi_k, \tag{10}$$

I_2 is MAI component from the users of other B spot beams, and given by

$$I_2 = \sum_{k=K+1}^{(B+1)K} \sqrt{2P}R_k S_k \beta_k Y_k \cos\phi_k, \tag{11}$$

and N is AWGN component, and given by

$$N = \int_0^T l(t)n(t)dt. \tag{12}$$

In (10) and (11), Y_k is cross-correlation component, and defined by

$$Y_k = \int_0^T d_k(t-\tau_k)c_k(t-\tau_k)c_1(t)dt. \tag{13}$$

The MAI in a DS/CDMA system is the most dominant factor in limitating system capacity. Since *p.d.f.* of MAI and self interference is difficult to obtain, the DS/CDMA system analysis has been evaluated using approximations [14,15]. For a large number of users and large processing gain, the *p.d.f.* is usually assumed to approximate Gaussian process with a variance equal to the sum of variances of individual interfering users. When the MAI is modeled as jointly Gaussian random variables, then the MAI components are unconditionally independent. Thus the effect of MAI is incorporated into analysis of BER performance simply by increasing variance of interference.

The total variance of interference and noise is given by

$$\sigma_t^2 = \sigma_{I_1}^2 + \sigma_{I_2}^2 + \sigma_N^2, \tag{14}$$

where $\sigma_{I_1}^2$ is variance of I_1, and given by

$$\sigma_{I_1}^2 = E[R_k^2 S_k^2]\frac{2PT^2}{3L}(K-1), \tag{15}$$

$\sigma_{I_2}^2$ is variance of I_2, and given by

$$\sigma_{I_2}^2 = E[R_k^2 S_k^2]\frac{2PT^2}{3L}\sum_{k=K+1}^{(B+1)K}\beta_k^2, \tag{16}$$

and σ_N^2 is variance of N, and given by

$$\sigma_N^2 = \frac{N_0 T}{2}. \tag{17}$$

For simplicity of analysis, it is assumed that unshadowed average received power is equal for each user of every spot beam. That is, $\mathrm{E}[R_k^2 S_k^2] = \mathrm{E}[R^2 S^2]$ for all $k = 1, 2, \cdots, K, \cdots, (B+1)M$. In (15) and (16), λ is defined by

$$E[R^2 S^2] = (1-z)\hat{p} + z\hat{p}\exp(hm_s + \frac{h\sigma_s^2}{2}), \tag{18}$$

where $\hat{p}$ is unshadowed average received power and $z(0 \le z \le 1)$ is probability that the satellite channel is in a shadowed Rayleigh fading state.

For notational simplicity, the subscript k in ϕ_k is omitted in the following analysis. Then, the conditional probability of bit error conditioned on fading statistics is given by

$$P_b(\phi|p) = Q(\frac{\sqrt{E_b}}{\sigma_t}\cos\phi), \tag{19}$$

where $Q(x) = \int_x^\infty \frac{1}{\sqrt{2\pi}}e^{-t^2/2}dt$ is Gaussian tail distribution and $E_b = PT$ is energy per bit.

For carrier phase tracking, a basic PLL structure is used in the system model. The linear analysis of PLL produces large error between analytical results and actual performance for phase locking in a low SNR region. Thus, a nonlinear analysis based on Fokker-Planck method is employed in deriving steady-state *p.d.f.* of phase noise For the first-order PLL, the transfer function of loop filter is $F(s) = 1$.

The steady-state *p.d.f.* of phase error for the first-order PLL is given by [16]

$$\frac{\partial}{\partial\phi}[\eta_1(\phi)p(\phi)] = \frac{1}{2}\frac{\partial}{\partial\phi^2}[\eta_2(\phi)p(\phi)], \tag{20}$$

where $\eta_i(\phi)(i = 1, 2)$ is defined by

$$\eta_i(\phi) = \lim_{\Delta t\to 0}\frac{1}{\Delta t}E[(\Delta\phi)^i|\phi]. \tag{21}$$

By solving the above equation (20), we can get the *p.d.f.* of phase noise, called Tikhonov distribution, given by

$$p_s(\phi) = \frac{\exp(\alpha_s\cos\phi)}{2\pi I_0(\alpha_s)}, \quad |\phi| \le \pi, \tag{22}$$

where ϕ is phase error, $I_0(x)$ is zeroth-order Bessel function of first kind, and α_s is loop SNR of tracking loop. When α_s is large, $p(\phi)$ decays very rapidly, so $p(\phi)$ becomes very small for all but small values of ϕ [17]. The probability distribution for phase noise is very important in that it directly influences detector performance.

In the AWGN channel, the loop SNR is basically given by signal power divided by loop noise bandwith times two-sided power spectral density of noise. The MAI component should be incorporated in evaluating loop SNR in a satellite channel for a DS/CDMA system. Therefore, the loop SNR is modified by

$$\alpha_s = \frac{P}{B_L \sigma_t^2}, \tag{23}$$

where B_L is loop bandwidth. It is known that for large loop SNR, the *p.d.f.* of phase noise is approximated as Gaussian distribution, and the variance of phase noise, σ_{ϕ^2}, is given by α_s^{-1}.

Then, the variance of phase noise is given by

$$\sigma_\phi^2 = \int_{-\pi}^{\pi} \phi^2 p(\phi) d\phi \tag{24}$$

Then, the average bit error probability over phase noise distribution is given by

$$P_{avg,b} = \int_{-\pi}^{\pi} P_b(\phi) p_s(\phi) d\phi. \tag{25}$$

IV. SIMULATION RESULTS

When the circulator reflector-type antenna is used, the antenna pattern is given by

$$G(x) = \left(\frac{2I_1(x)}{x}\right), \tag{26}$$

where $I_1(x)$ is the first-order Bessel function of the first kind, $x = \pi d \sin\phi_j/\lambda$, ϕ_j is antenna pointing angle, d is antenna diameter, λ is wavelength. The value of d and λ depend on the half power beam-width which equals to $1.02d/\lambda$.

For simulation examples, the carrier frequency = 1.625GHz (L-band), satellite altitude = 10354km (MEO (medium earth orbit) satellite), half power beam-width = 6.3^o, ratio of loop bandwidth and data bit rate B_L/R = 0.1, channel state parameter $z = 0.3$, Rician factor R_f = 6 dB, and processing gain of spreading sequence L = 128 are assumed. For simplicity, it is assumed that the effect of MAI from all spot beams other than reference beam contributes equal amount of MAI as due to users in the reference beam.

In Fig. 1, variance of phase noise vs. SNR is shown for varying number of users in a beam. The simulation examples are for elevation angle = 40 degree, and σ_s = 4 dB. It is shown that the variance of phase noise is significantly influenced by MAI which is inherent in a CDMA system. Therefore, it is confirmed that MAI should be considered in determining phase noise distribution, eventually limiting

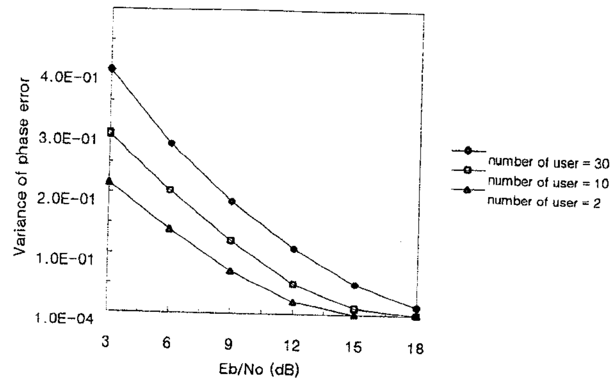

Fig. 1. Variance of phase noise vs. SNR.

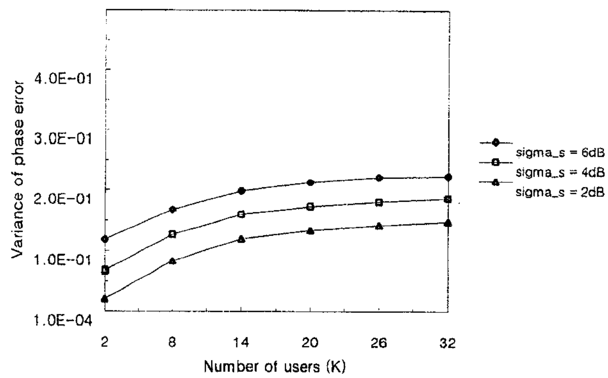

Fig. 2. Variance of phase noise vs. the number of users in a beam.

the selection of loop bandwidth. The difference of variance is more prominent for the low SNR case than for the high SNR case because the low SNR case is more sensitive to detection loss than the high SNR case.

In Fig. 2, variance of phase noise vs. the number of users in a beam is shown for varying standard deviation of shadowing. The simulation examples are for SNR = 10dB and elevation angle = 40 degree. It is demonstrated that the heavy shadowing derives larger variance of phase noise, which results in higher BER. And, the degree of performance degradation is almost uniform for all the number of users because shadowing affects all the users in a beam with almost the same level.

In Fig. 3, bit error probability vs. SNR is shown for varying phase errors. The simulation examples are for elevation angle = 40 degree, σ_s = 4 dB, and K=10. When the phase noise exists, the BER performance is degraded for both low and high SNR cases. This can be interpreted that the phase noise incurs decrease of effective SNR, which leads to detection loss. The performance degradation due to phase noise is more significant for the low SNR case.

In Fig. 4, bit error probability vs. standard deviation of phase noise, σ_ϕ, is shown for varying number of users. The simulation examples are for SNR = 10dB, elevation angle = 40 degree, and σ_s = 4 dB. It is shown that the bit error probability increases as the nubmer of users because an

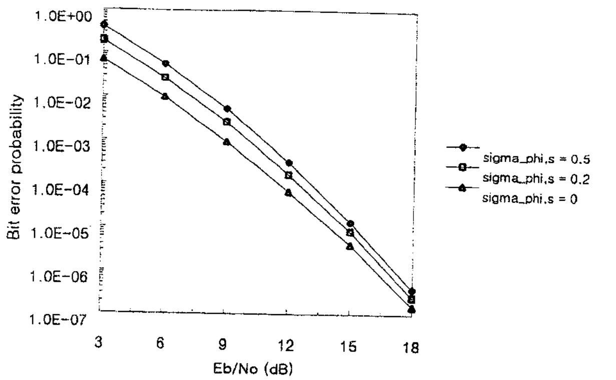

Fig. 3. Bit error probability vs. SNR for different phase errors.

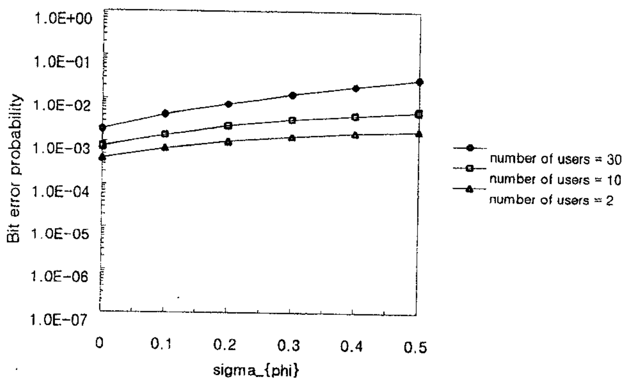

Fig. 4. Bit error probability vs. σ_ϕ for different number of users.

increase of the number of users incurs an increase of MAI. For the case of large number of users, the performance difference between the cases with and without phase noise becomes larger because of the effect of increased MAI.

In Fig. 5, bit error probability vs. standard deviation of phase noise is shown for varying standard deviations of shadowing. The simulation examples are for SNR = 10dB, $K = 10$, and elevation angle = 40 degree. It is shown that the bit error probability increases with the standard deviation of shadowing. Thus, it can be noted that shadowing also has a significant impact on the BER performance in a satellite channel.

V. CONCLUSIONS

The impact of phase noise on a DS/CDMA system was analyzed and simulated in a satellite channel. The nonlinear Fokker-Planck method was employed to analyze the first-order PLL performance. The simulation results for BER performance with phase noise were shown in terms of many system parametes such as SNR, the number of users, and standard deviation of shadowing. From the simulation results, it was demonstrated that the performance of a CDMA-based satellite system is substantially degraded due to phase noise. This can be understood that the BER performance degradation is caused by phase noise which

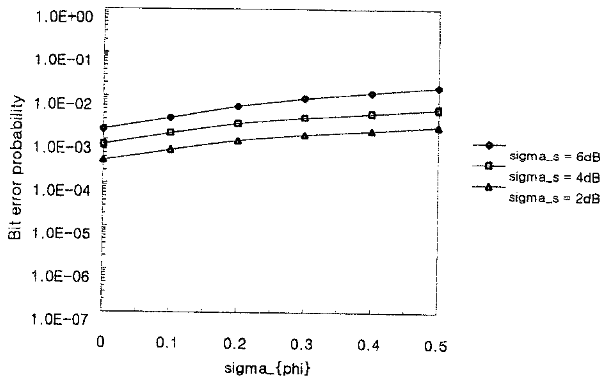

Fig. 5. Bit error probability vs. σ_ϕ for different level of shadowing.

lowers effective SNR leading to detection loss. The analysis in this paper is concentrated on the first-order PLL. However, the methodology in this paper can be applicable to higher-order PLL in a straight forward manner. The results in this paper can be applied to the satellite link design for a CDMA-based satellite system.

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